VARUN KANULLA

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PROFESSIONAL SUMMARY

Data Scientist with 4+ years of experience working on production-level machine learning and AI solutions in healthcare and SaaS environments. Focused on solving real business problems using data, with hands-on experience in predictive modeling, NLP, and LLM-based systems. Known for building practical solutions that reduce manual effort, improve decision-making, and scale across large user bases. Comfortable working across data pipelines, model development, and deployment in cloud environments.

SKILLS

Machine Learning & Statistics: Supervised Learning, Unsupervised Learning, Predictive Modeling, Feature Engineering, Model Evaluation, Hyperparameter Tuning, A/B Testing, Hypothesis Testing, Survival Analysis (Cox Models, Random Survival Forests)
AI & NLP: Natural Language Processing (NLP), Named Entity Recognition (NER), Relation Extraction, Text Classification, Sentiment Analysis, Social Determinants of Health (SDoH) Extraction
Generative AI & LLMs: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Evaluation, Semantic Search, Vector Databases (FAISS, Pinecone)
Programming & Libraries: Python, SQL, Pandas, NumPy, Scikit-learn, XGBoost, PySpark, TensorFlow (basic), R (statistical modeling, clustering, survival analysis)
Big Data & Distributed Systems: Apache Spark, Hadoop, Kafka, Hive
Data Engineering & Pipelines: ETL Pipelines, Data Cleaning, Data Transformation, Data Validation, Feature Pipelines, Data Integration (multi-source systems, EHR/CRM data)
Frameworks & APIs: FastAPI, Flask, REST APIs, Microservices, Real-Time Inference Pipelines
Cloud & MLOps: AWS (S3, Lambda, SageMaker), Docker, Kubernetes, MLflow, CI/CD Pipelines, Model Deployment, Model Monitoring
Visualization & Reporting: Power BI, Tableau, Matplotlib, Seaborn, Data Storytelling
Databases & Tools: PostgreSQL, MongoDB, Git, Jupyter Notebook, ABC CRM
Core Competencies: Analytical Thinking, Problem Solving, Stakeholder Communication, Cross-Functional Collaboration

CERTIFICATIONS

Fundamentals of Accelerated Computing with CUDA Python
Fundamentals of Deep Learning
AWS Academy Graduate - AWS Academy Machine Learning

EXPERIENCE

Cigna Healthcare, USA | AI Engineer — Jan 2025 – Current

- Designed and implemented large-scale clinical NLP pipelines processing over 400K de-identified notes across 4,000+ patients, extracting Social Determinants of Health and comorbidity signals for oncology research.
- Built end-to-end data workflows converting raw EHR data into structured patient-level datasets, linking note-level records and handling duplicates, missing values, and multi-source clinical data integration.
- Implemented and benchmarked multiple SDoH extraction approaches, including MedSDoH, BERT-based NER models, and LLM pipelines, evaluating trade-offs across accuracy, scalability, and computational efficiency.
- Engineered preprocessing pipelines to transform large clinical datasets into model-ready inputs, reducing nearly 500K notes to 120K relevant records through domain-specific filtering and text normalization techniques.
- Developed patient-level feature engineering pipelines generating structured SDoH variables across 20+ domains, supporting downstream epidemiological analysis and improving interpretability of clinical insights.
- Conducted multimorbidity and comorbidity analysis using hierarchical clustering techniques, identifying disease co-occurrence patterns across oncology cohorts and supporting clinical research studies.
- Built survival analysis models using Cox Proportional Hazards and Random Survival Forests, achieving a C-index of 0.82 for predicting long-term patient outcomes across cancer populations.
- Designed evaluation frameworks combining precision-recall metrics and manual validation workflows, improving reliability and consistency of NLP-based clinical data extraction pipelines.
- Developed LLM-based classification pipelines for clinical note interpretation, generating structured outputs and benchmarking multiple open-source models for performance and accuracy across datasets.
- Optimized LLM inference workflows by implementing batching, truncation, and chunking strategies, reducing computational cost and improving processing efficiency for large-scale clinical text analysis.
- Built standardized benchmarking pipelines evaluating models such as LLaMA, Mistral, and Gemma, enabling reproducible experimentation with schema-constrained outputs across multiple datasets.
- Processed long-form clinical text using chunking strategies to handle documents up to 135K characters, ensuring compatibility with model context limits and improving inference stability.
- Developed reproducible machine learning workflows using Python and R, ensuring experiment traceability, version control, and generation of analysis outputs suitable for research publications.
- Collaborated with clinical researchers and domain experts to translate epidemiological requirements into scalable AI solutions, supporting IRB-approved studies and improving data usability for healthcare research.

Freshworks Inc, India | Data Scientist **Sep 2020 – Nov 2023**

- Built customer churn prediction models using XGBoost and logistic regression on large-scale product usage and billing datasets, improving retention targeting and reducing churn across SaaS platforms by 14%.
- Designed and maintained end-to-end ETL pipelines using Python and SQL, processing high-volume customer interaction data and improving data availability for analytics and machine learning workflows.
- Developed NLP pipelines to classify and prioritize customer support tickets, improving issue routing accuracy and reducing resolution time across support operations by 26%.
- Engineered behavioral and engagement-based features from product usage logs, improving model performance and enabling deeper insights into customer lifecycle and product adoption patterns.
- Created real-time dashboards using Power BI and Tableau to track key performance indicators, enabling stakeholders to monitor user engagement, churn trends, and product performance metrics effectively.
- Deployed machine learning models using REST APIs with FastAPI, ensuring seamless integration into backend systems and enabling real-time predictions across customer-facing applications.
- Conducted A/B testing and statistical analysis on product features, measuring impact on user engagement and supporting data-driven decisions for product improvements and releases.
- Worked with large datasets using Apache Spark, improving data processing efficiency and enabling scalable analytics across distributed environments handling high-volume SaaS data.
- Built recommendation systems using collaborative filtering techniques, improving user engagement and increasing feature adoption across platform users by 22%.
- Collaborated with product managers and engineering teams to translate business requirements into data-driven solutions, ensuring alignment with product goals and successful deployment of analytics initiatives.

EDUCATION

Masters in CISE (Computer & Information Science & Engineering) | University of Florida, Gainesville, Florida USA
Bachelors in Computer Science with AI Specialization | Bennett University, Greater Noida, India

PROJECTS

US Domestic Flight Delay Prediction

- Trained and compared 10+ classifiers including XGBoost, GBM, Random Forest, Neural Networks, and stacked ensembles using 5-fold cross-validation.
- Built compare Carrier Delays ( ) function to compare predicted delay probabilities across all carriers for any user-specified route and date.
- Key predictors identified: time of day, carrier, origin/destination state, and day of week.
- Full pipeline in R: preprocessing, feature engineering, resampling, training, and model stacking.
- XGBoost achieved best balanced accuracy; stacked model (GBM + XGBoost → regularised logistic meta-learner) correctly classifies 62.2% of flights.

Telecom Churn Intelligence Platform

- Full pipeline: data ingestion (CSV + Hugging Face API) → validation → preparation → feature engineering → feature store → DVC versioning → model training.
- Supports Random Forest and Logistic Regression classifiers with class-balanced training; all experiments tracked via MLflow.
- DVC-based versioning creates reproducible snapshots at every pipeline stage.
- Orchestrated with Apache Airflow; containerized with Docker and Docker Compose.
- Centralized logging system with per-stage log files; targets >85% accuracy and >0.80 F1-score.